

Intraday Tick Simulator

Mathematical Foundations

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di.simtick and **di.simmarket** modules for KDB-X

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Abstract

This document gives the mathematical framework of a simulator that generates synthetic but realistic intraday market data (trades and quotes) for several stocks over a calendar of trading days.

Trade and quote arrivals are self-exciting Hawkes processes, simulated without approximation. The mid price follows a geometric Brownian motion with optional jumps; its variance accrues with trading activity, and signed order flow moves it through market impact. Quotes are generated first, with spreads in whole ticks, and each trade executes against the quote in force, with buyer- and seller-initiated trades arriving in runs.

Consecutive days are linked by overnight price gaps. Volatility and volume vary from day to day, in quiet and busy periods that last several days, and busy days tend to be volatile ones; any single day can be regenerated on its own.

Stocks move together through common factors and common jumps, with loadings that leave each stock's volatility unchanged and imply a correlation matrix that is valid by construction. A market may trade in two sessions around a mid-day break.

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1 Introduction

A realistic market simulator should reproduce the main features of real intraday data:

- **Trade clustering:** trades arrive in bursts, and quote updates follow them.
- **Intraday pattern:** trading is busy in the first half hour, quiet at midday and busiest before the close; volatility follows the same shape; spreads are widest just after the open and narrow within the first half hour.
- **Price moves:** prices move continuously, with occasional sudden jumps, and a jump is followed by a burst of trading.
- **Order-flow impact:** a buy pushes the price up and a sell pushes it down, and part of the move fades afterwards. Because buys tend to follow buys and sells follow sells, these small moves add up.
- **Quotes and trades:** quotes come first and trades happen against them; the spread is a whole number of ticks, usually one; most trades happen at the bid or the ask, some at the midpoint or slightly inside; trade sizes mix round lots of 100 shares with smaller odd lots; and every trade records its venue, a condition code and a sequence number.
- **Days:** part of each day’s price movement happens overnight, between the close and the next open; volatility and volume go through quiet and busy periods lasting several days; and the calendar has holidays and half days.
- **Co-movement:** stocks move together, more so within a sector, and market-wide news moves all of them at the same instant.

The simulator is made of two KDB-X modules. **di.simtick** generates one trading day for one instrument. **di.simmarket** runs it for several instruments over a calendar of trading days, linking each day to the next, and writes the result as a date-partitioned database. The configuration of a run is composed by **di.simtick.config**, a child module of **di.simtick** that both use, from four layers: the market, the instrument, the scenario and the run.

Each model in the following sections is given with the configuration settings that control it, written like **vol** or **spreadticks**. All random numbers come from seeded generators, so the same configuration always produces the same data.

2 Random Numbers and Seeds

Standard normal variates come from the Box-Muller transform. Let U_1, U_2 be independent uniforms on $(0, 1)$; then

$$R = \sqrt{-2 \ln U_1}, \quad \Theta = 2\pi U_2, \quad Z_1 = R \cos \Theta, \quad Z_2 = R \sin \Theta \quad (1)$$

are independent standard normals. The implementation draws $1 - U$ so that the argument of the logarithm is never 0. Poisson variates with element-wise means λ_i are drawn by inversion: with U_i uniform, X_i is the number of $k \geq 0$ with $F(k; \lambda_i) < U_i$, the cumulative distribution accumulated term by term and truncated at a value beyond which the tail is negligible for the means in use.

A day’s stream is seeded once, from the configuration’s **seed**. In a multi-day run the seeds of a date are derived from the run’s seed, the date and the ticker, and how they are derived matters. The generator gives correlated streams for seeds that are related: two seeds a constant apart, or consecutive, give draws that correlate at 0.7 and more all along the stream, 0.96 at the 2000th draw. Seeds computed by arithmetic from the date and the ticker are related in exactly

that way, between instruments on a date and between consecutive dates, so every seed is a hash of what it depends on,

$$H(a, b) = 1 + \left(\text{the first four bytes of md5}(a \parallel b) \right) \bmod (2^{31} - 2) \quad (2)$$

Every date d receives a *regime seed* from the seed s and the date alone, shared by every instrument run on that date (the market's day), and from it the instrument's *day seed* and *gap seed*, and the *factor seed* of Section 8:

$$\begin{aligned} s_{\text{regime}} &= H(s, \text{days}(d)), & s_{\text{day}} &= H(s_{\text{regime}}, h(\text{sym})), \\ s_{\text{gap}} &= H(s_{\text{day}}, 3), & s_{\text{factor}} &= H(s_{\text{regime}}, 5) \end{aligned} \quad (3)$$

with h a hash of the ticker's characters. A date's regime innovation is therefore the same in any calendar that contains it, any day's tape can be regenerated alone from its row of the summary table, and the streams of two instruments, or of two dates, are independent: over eight years of dates the first draws of two instruments' streams correlate within 0.03, as do those of consecutive dates.

3 Arrivals: Hawkes Processes on Two Clocks

3.1 Model

Trade arrivals exhibit self-excitation: each trade raises the probability of the next.

Definition 1 (Hawkes process). *A univariate Hawkes process $N(t)$ is a counting process with conditional intensity*

$$\lambda(t) = \lambda_0(t) + \sum_{t_i < t} \alpha e^{-\beta(t-t_i)} \quad (4)$$

where $\lambda_0(t)$ is the baseline intensity, $\alpha > 0$ the excitation per event (**alpha**), $\beta > 0$ the decay (**beta**) and $\{t_i\}$ the event times before t .

Proposition 1 (Stationarity). *The process is stationary if and only if the branching ratio $n = \alpha/\beta$ is below 1. n is the expected number of events each event triggers; the mean intensity is $\lambda_0/(1 - n)$.*

3.2 Intraday Profile

The baseline follows the session's profile:

$$\lambda_0(t) = \mu s\left(\frac{t}{T}\right) \quad (5)$$

where μ is **baseintensity**, T the session length in seconds and s a shape function. The shape is a list of positive weights $w_0, \dots, w_{n-1}$, one per equal bin of the session (**profile**, thirteen half hours in the shipped market file), interpolated linearly between bin midpoints and flat beyond the first and last:

$$s(p) = w_i + (x - i)(w_{i+1} - w_i), \quad x = \min\left(n - 1, \max\left(0, np - \frac{1}{2}\right)\right), \quad i = \min(n - 2, \lfloor x \rfloor) \quad (6)$$

A binned profile expresses a flat midday and a spike in the last minutes, which the earlier three-multiplier cosine could not. The shape never exceeds $\max_i w_i$, which the simulation relies on.

3.3 Sessions and the Mid-Day Break

A market may have one mid-day break $[t_b, t_e]$ (**breakstart**, **breakend**), as the Singapore and Hong Kong exchanges do. The simulator then works in trading time, the break removed: with T the trading seconds of the day, the arrival processes, the profile $s(\cdot)$ and the clocks of Section 4 live on $[0, T)$, and a trading time u is mapped to the wall clock by $u \mapsto u + (t_e - t_b) \mathbf{1}\{u \geq t_b\}$, so no trade or quote falls in the break and the last quote stays in force through it. The mid takes a gap over the break,

$$\ln \frac{S(t_e)}{S(t_b^-)} \sim \mathcal{N}\left(-\frac{1}{2}v_b, v_b\right), \quad v_b = \text{breakshare} \frac{\sigma^2}{D}, \quad (7)$$

with D the trading days of the year, while the diffusion runs at $\sigma\sqrt{1 - \text{breakshare}}$, so the variance of the day, and with the overnight gap of Section 7 the close-to-close variance, is unchanged. A reopening print at t_e (condition code B) prints a share **breakauctionpct** of the continuous volume at the afternoon's first mid. A half day on such a market is the morning session alone. The auction phases of the exchanges are single prints, there is no pre-market or after-hours trading, and times are the exchange's local time.

3.4 Exact Simulation by the Cluster Representation

Ogata's thinning simulates a Hawkes process by accepting candidates of a Poisson process at a rate that must bound the intensity. With a fixed bound the acceptance probability is capped at 1 in bursts, which under-produces arrivals: 5% at branching ratio 0.4 and threefold at 0.9 in the earlier implementation. The simulator now uses the cluster representation of Hawkes and Oakes (1974), which is exact and needs no bound: immigrants arrive as an inhomogeneous Poisson process of intensity $\lambda_0(t)$, and every event spawns Poisson(n) children at $\text{Exp}(\beta)$ delays, generation after generation until a generation is empty. The union of all generations is the process.

Algorithm 1: Hawkes process by clusters

```

Input: mu, alpha, beta, session length T, shape s, extra immigrant times E
Output: event times
1: cand <- homogeneous Poisson process on [0,T) at rate mu*max(s)
   (exponential waits drawn in blocks until past T)
2: immigrants <- {c in cand : U_c < s(c/T)/max(s)} union E    # exact thinning
3: parents <- immigrants; events <- immigrants
4: while parents is not empty do
5:   k <- Poisson(|parents| * alpha/beta)                        # children in total
6:   children <- parents[uniform indices of size k] + Exp(beta) delays
7:   children <- {t in children : t < T}
8:   events <- events union children; parents <- children
9: end while
10: return sorted events

```

Line 5 draws the total number of children of a generation and line 6 assigns each child a parent uniformly, which has the same law as Poisson(n) children per parent and is one vector operation. Counts match the analytic mean $T\mu/(1 - n)$ within 1% at every branching ratio tried, up to 0.95, and a day of 500,000 trades takes tens of milliseconds.

3.5 Jumps Seed Bursts

A price jump (Section 4.2) brings a burst of activity that fades over minutes. Each jump at time t_j adds **jumpburst** extra immigrants at exponential delays of mean **jumpburstminutes**, each

with its usual cascade, so the burst carries $\text{jumpburst}/(1 - n)$ trades in all. The quote clock receives the same shock scaled by quotespertrade . The common jumps of Section 8 seed the same bursts on every stock that takes them, and the trades the bursts are expected to add are taken out of the base intensity, so the day still averages tradesperday trades.

3.6 The Quote Clock

Quote updates form a second Hawkes process with the same α , β and profile. Its immigrants come from two sources: a background clock at intensity $(1 - \rho_q) q \mu s(t/T)$, and, after each trade, $\text{Poisson}(\rho_q q (1 - n))$ immigrants at $\text{Exp}(\beta)$ delays, where q is quotespertrade and ρ_q is quotetradelink . The expected number of quote updates per trade is q , and a share ρ_q of them follows the trades, so a burst of trades brings a burst of quotes: with flat profiles the per-second counts of the two clocks correlate at 0.38 against 0 without the link. A quote is placed at the open before all others, so a quote is in force for every trade.

4 The Mid: Price Path, Clock and Order-Flow Impact

4.1 Geometric Brownian Motion

Definition 2 (GBM). *The price S_t satisfies $dS_t = \mu S_t dt + \sigma S_t dW_t$, with drift μ (*drift*), volatility σ (*vol*), both annualized, and W a standard Brownian motion.*

By Itô's lemma, over a sequence of points $t_0 = 0 < t_1 < \dots < t_m$ with steps Δ_j in years,

$$S_{t_i} = S_0 \prod_{j=1}^i \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) \Delta_j + \sigma \sqrt{\Delta_j} \varepsilon_j \right], \quad \varepsilon_j \sim \mathcal{N}(0, 1) \quad (8)$$

The path is sampled on the quote clock, whose first point is the open itself, so S_0 is the price at the open (*price*). The mid at a quote is the path there.

4.2 Jumps as Events

Under $\text{pricemodel} = \text{jump}$, the day's jumps are events: their number is $\text{Poisson}(\text{jumpintensity})$, their times uniform in the session, their sizes lognormal,

$$J_k = \exp(\mu_J + \sigma_J \varepsilon_k), \quad S_t = S_t^{\text{gbm}} \prod_{t_k \leq t} J_k \quad (9)$$

with μ_J and σ_J the keys jumpmean and jumpvol . A jump is applied to the path from the first quote at or after it, and it seeds the bursts of Section 3.5.

4.3 Calendar and Transaction Clocks

The steps Δ_j depend on the *clock*. Under the calendar clock,

$$\Delta_j = \frac{t_j - t_{j-1}}{N_{\text{days}} T} \quad (10)$$

with N_{days} the trading days per year (tradingdays) and T the session in seconds: variance grows with time and realized volatility is flat through the day. Under the transaction clock, the default, every quote update carries the same variance, one trading day's over the day's m updates,

$$\Delta_j = \frac{1}{N_{\text{days}} m} \quad (11)$$

so variance grows with activity: realized volatility follows the intraday profile and rises in bursts. Because the day's variance is normalized by its own number of updates, a busier day does not carry more variance; activity only redistributes it within the day. On the shipped NVDA configuration, hourly realized volatility runs from 0.36 at 11:00 to 0.60 in the last hour while the day-level figure stays at the configured 0.45.

4.4 Order-Flow Impact

Signed trades move the mid. The order flow of the day, aggressor signs $\varepsilon_i = \pm 1$ and quantities Q_i (Section 6), is drawn before the quotes, and each trade contributes an impulse in ticks,

$$a_i = \kappa \delta \varepsilon_i \sqrt{\frac{Q_i}{\bar{Q}}} \quad (12)$$

where κ is `impactticks`, δ the tick size and $\bar{Q}$ the mean trade size of the quantity model. A share π (`impactpermanent`) of each impulse stays; the rest halves every h seconds (`impacthalflifeseconds`). The shift in force at a quote time t is

$$D(t) = \sum_{t_i < t} a_i \left[\pi + (1 - \pi) 2^{-(t-t_i)/h} \right] \quad (13)$$

computed as a cumulative sum for the permanent part and a scan of the transient part over the trades, and the mid becomes $S_t + D(t)$. With persistent signs the tape shows price impact and partial reversion: on the shipped NVDA configuration the signed markout is 0.34 cents one second after a trade and 0.17 cents after a minute, against 0.007 cents with $\kappa = 0$. Impact adds about 3% to day-level volatility.

5 Quotes

5.1 Spread in Ticks

The spread is a whole number of ticks: one tick plus a Poisson excess whose mean is the mean spread less one tick,

$$\text{spread}_t = \delta (1 + X_t), \quad X_t \sim \text{Poisson}(\max(0, \bar{s} g(t) a(t) - 1)) \quad (14)$$

with $\bar{s}$ the mean spread in ticks through the day (`spreadticks`, 1.15 for the shipped NVDA configuration), g the time-of-day multiplier and a the activity multiplier below. The bid is placed so that the spread sits around the mid, on the tick grid:

$$B_t = \delta \left\lfloor \frac{M_t - \text{spread}_t/2}{\delta} + \frac{1}{2} \right\rfloor, \quad A_t = B_t + \text{spread}_t \quad (15)$$

A large cap's spread is widest in the first minutes and tightens within the half hour; it is tightest at the close. The multiplier is the midday level moved toward the open and close levels by exponential decays of e-folding time τ minutes (`spreaddecayminutes`):

$$g(t) = g_{\text{mid}} + (g_{\text{open}} - g_{\text{mid}}) e^{-t/\tau} + (g_{\text{close}} - g_{\text{mid}}) e^{-(T-t)/\tau} \quad (16)$$

with the three levels `spreadopenmult`, `spreadmidmult` and `spreadclosemult`, 2.5, 1 and 0.9 in the shipped market file. On the shipped NVDA configuration the spread is one tick 81% of the time, with a midday mean of 1.15 ticks, 1.9 in the first half hour and 1.10 in the last.

5.2 Activity Widens the Spread

Bursts widen spreads. At each quote the number of quote updates in the trailing minute, c_t , is compared with the number the intensity profile expects there, e_t , and the mean spread is multiplied by

$$a(t) = \left(\text{clip} \left[\frac{1 + c_t}{1 + e_t}, \frac{1}{4}, 4 \right] \right)^\gamma \quad (17)$$

with γ the key `spreadactivity` (0.5 in the shipped market file; 0 turns the coupling off). In the minutes after a jump the spread is 1.3 to 2 times its median, and with flat profiles the per-minute spread tracks the per-minute quote count at a correlation of 0.58 against 0.05 without the coupling. Under the transaction clock this also couples spread to volatility, since both follow activity.

5.3 Sizes and Imbalance

Quote sizes are lognormal around `avgquotesize` with log-spread `quotesizevol`, in round lots of 100, and tilted toward the side of the next mid move: with ΔM the sign of the next change of the mid and θ the key `imbalance` signal,

$$\text{bidsize} \propto \exp(\ell z_b + \theta \text{sgn } \Delta M - \frac{1}{2} \ell^2), \quad \text{asksize} \propto \exp(\ell z_a - \theta \text{sgn } \Delta M - \frac{1}{2} \ell^2) \quad (18)$$

so the book leans, weakly, toward what comes next: at $\theta = 0.15$ the size imbalance correlates with the next move at 0.21, a regularity the order book shows, and 0 without the signal. About half of consecutive quotes change only in size, the cancels and replaces of a real feed.

6 Trades Against the Quote in Force

6.1 Aggressor Side

Each trade arrival takes the quote in force at its time. Its aggressor side follows a Markov chain: the first sign is uniform, and each later sign repeats the previous one with probability π_s (`sidepersistence`), so that

$$\text{corr}(\varepsilon_i, \varepsilon_{i-1}) = 2\pi_s - 1 \quad (19)$$

0.4 in the shipped market file, in the range measured on equities. Trades carry the side in an `aggressor` column (B or S), which makes effective spread, realized spread, Lee-Ready classification and markouts well defined.

6.2 Where a Trade Prints

A buyer-initiated trade prints at the ask and a seller-initiated one at the bid, except for a share `midpointshare` that prints at the midpoint and a share `improvementshare` that prints a tenth of a tick inside the touch (price improvement). Prices sit on the tenth-of-a-tick grid. On the presets 80% of trades print at the touch, 12% at the midpoint and 8% inside, the effective spread is 0.5 basis points on NVDA and no trade prints outside its prevailing quote.

6.3 Quantities

Under `qtypemodel = mixture`, a trade's size is a round lot with probability `roundlotshare` (100, 200, 300, 500 or 1000 shares with weights 0.5, 0.2, 0.1, 0.12, 0.08), a block with probability `blockshare` (lognormal, median `blockqty`, log-spread 0.5), and otherwise an irregular lot,

$$Q = \max(1, \lfloor e^X \rfloor), \quad X \sim \mathcal{N}(\ln \bar{Q} - \frac{1}{2} \sigma_Q^2, \sigma_Q^2) \quad (20)$$

with mean $\bar{Q}$ (`avgqty`) and log-spread σ_Q (`qtyvol`). The lognormal and constant models remain. On the shipped NVDA configuration 35% of trades are round lots and 55% odd lots, the mean size 153 shares. The mean size of the mixture, which scales the impact of Section 4.4, is $(1 - r - b)\bar{Q} + 260r + be^{1/8}\text{blockqty}$.

6.4 Tape Attributes and Auctions

Each trade carries a condition (`cond`: R regular, I odd lot, below the smallest round lot, O and C the auction prints, B the reopening print after a mid-day break), a venue and a sequence number. Midpoint and improved prints are off-exchange with probability 0.95 (venue TRF); prints at the touch are split so that the day's off-exchange share is `offexchangeshare`; lit prints are spread over the MIC codes XNAS, XNYS, ARCX, BATS, EDGX and IEXG by share. One sequence runs across quotes and trades in time order, the opening quote first. The opening and closing auction prints frame the session: at the first and last mid, for `openauctionpct` and `closeauctionpct` of the continuous volume, on the primary listing venue, with no aggressor.

7 Days: Overnight Gaps, Regimes and Calendars

7.1 Overnight Gap and the Variance Budget

Each day after the first opens at the previous close moved by an overnight log return, normal with no drift (the open is a martingale of the close) and variance a share ω (`overnightshare`) of one trading day's, weighted for the calendar days of the gap by κ_g (`gapdayweight`):

$$r_{\text{on}} \sim \mathcal{N}\left(-\frac{1}{2}v, v\right), \quad v = \omega \frac{\sigma^2}{N_{\text{days}}} (1 + \kappa_g(\Delta d - 1)) \quad (21)$$

The intraday simulation runs at $\sigma(1 - \omega)^{1/2}$, so over a one-night gap the close-to-close variance is exactly σ^2/N_{days} : the configured volatility is the close-to-close one, as quoted. A weekend carries more than a night and less than three. With factor loadings the standard normal behind the gap mixes the factors' overnight normals of the date with the stock's own (Section 8), so the gaps of two stocks correlate as their days do. Over six months of NYSE days the measured overnight share of close-to-close variance is 29% for $\omega = 0.3$.

7.2 Day-Level Regimes

Days differ, in spells. Two standardized AR(1) states with persistence ϕ (`regimepersistence`), x for volatility and y for volume, are driven by daily shocks with correlation ρ (`regimecorr`):

$$x_d = \phi x_{d-1} + \sqrt{1 - \phi^2} \eta_d, \quad y_d = \phi y_{d-1} + \sqrt{1 - \phi^2} \xi_d, \quad \xi_d = \rho \eta_d + \sqrt{1 - \rho^2} \zeta_d \quad (22)$$

with η_d, ζ_d the two normals drawn from the date's regime seed. The day's multipliers are

$$m_d^{\text{vol}} = \exp(s_v x_d - s_v^2), \quad m_d^{\text{volume}} = \exp(s_q y_d - \frac{1}{2}s_q^2) \quad (23)$$

with s_v and s_q the keys `volregimesd` and `volumeregimesd`. The volume multiplier scales the base intensity and averages 1. The volatility multiplier scales the intraday σ , and since volatility enters a day as variance it is normalized on its square, $\mathbb{E}[(m^{\text{vol}})^2] = 1$, so that the mean daily variance is the configured one: the earlier normalization on the mean inflated close-to-close volatility by $\exp(s_v^2)$. Under the transaction clock a volume regime does not raise volatility (the spread of daily realized volatility is unchanged with the volume regime alone), so the two channels are separate and their correlation is ρ , 0.7 in the normal scenario. The calendar's own per-day multipliers, closing time and jump intensity multiply on top, for event days and half days.

7.3 Calendar

The calendar is a table of dates with optional closing time, multipliers and jump intensity. A generator produces the NYSE trading days of a range under its rules: weekdays less New Year's Day (not observed on the Friday when it falls on a Saturday), Martin Luther King Jr. Day, Presidents' Day, Good Friday from the Gregorian Easter, Memorial Day, Juneteenth from 2022, Independence Day, Labor Day, Thanksgiving and Christmas, Saturday holidays observed on the Friday and Sunday ones on the Monday, with early closes at 13:00 on the day after Thanksgiving, July 3 and Christmas Eve when they are trading days. 2026 has 251 trading days.

7.4 The Output Database

Several stocks over a calendar are written as a standard date-partitioned kdb+ database, one day of all stocks at a time: for each date in order every stock is simulated for that date, carrying its own previous close into the open, the stocks' tables are joined and the date's partition is written in one step (the trades, the quotes when requested, and a `days` table with one row per stock), sorted by symbol then time with the symbol column parted and enumerated against the root symbol file, every column compressed (128KB blocks, zstd level 3 by default), and the day's tables are dropped. Memory holds one day of all stocks; only each stock's close is carried forward. `days` is written last, so a crash cannot leave a date looking complete: a complete date is skipped on a rerun and an incomplete one rewritten from scratch, and since every date has its own seeds an interrupted run resumed equals a full run exactly. The run itself (every stock's composed configuration, the calendar, the tables, the compression, the module version and the git commit of the code) is stored at the root as the q object `simrun`, so the database is reproducible from that file alone with the same code, and a database built with another configuration is refused rather than mixed.

8 Co-movement: Common Factors and Common Jumps

8.1 Factors and Loadings

A market names K common factors (`factors`): a market factor, sectors, or statistical factors, any number. Each factor F_k is a Brownian motion in trading time, independent of the others, whose variance follows an intraday profile (`factorprofile`) and is 1 over the day: with w the profile normalized to integrate to 1 over the trading seconds,

$$\text{Var } F_k(u) = W(u) = \int_0^u w(v) dv, \quad W(T) = 1 \quad (24)$$

A stock i has loadings $b_i \in \mathbb{R}^K$ (`factorloadings`) with $|b_i|^2 \leq 1$. Its diffusion is split between the factors and its own Brownian motion W_i , which runs on the stock's clock (Section 4.3) with variance 1 over the day:

$$d \ln S_i = \frac{\sigma_i^d}{\sqrt{N_{\text{days}}}} \left(b_i \cdot dF + \sqrt{1 - |b_i|^2} dW_i \right) + \text{drift} \quad (25)$$

with σ_i^d the volatility of the day's diffusion. The variance of the day is unchanged, a loading is the stock's correlation with the factor, and the diffusive returns of two stocks correlate at

$$\rho_{ij}^{\text{diff}} = b_i \cdot b_j = \sum_k b_{ik} b_{jk} \quad (26)$$

The correlation matrix is $BB^\top + \text{diag}(1 - |b_i|^2)$, positive semi-definite for any loadings that satisfy $|b_i|^2 \leq 1$: nK numbers instead of $n(n-1)/2$, and never a matrix that fails to factorize.

On the path, the factors are evaluated on a one-second grid and the mid at a quote time u_j is multiplied by

$$\exp \left[\frac{\sigma_i^d}{\sqrt{N_{\text{days}}}} b_i \cdot F(u_j) - \frac{1}{2} \frac{(\sigma_i^d)^2 |b_i|^2}{N_{\text{days}}} W(u_j) \right] \quad (27)$$

while the stock's own part is the path of Section 4 at volatility $\sigma_i^d(1 - |b_i|^2)^{1/2}$. The standard normal behind the mid-day break's gap and the overnight gap is mixed the same way, $b_i \cdot G + (1 - |b_i|^2)^{1/2} z_i$, with G the factors' gap normals of the date.

8.2 Common Jumps and the Variance Budget

A factor can jump: the number of jumps of factor k in a day is $\text{Poisson}(\lambda_k)$ (`factorjumpintensities`), their times uniform in the trading time, their log sizes $Y \sim \mathcal{N}(0, s_k^2)$ (`factorjumpvols`). Every stock takes the jump at the same instant, times its jump loading ℓ_{ik} (`jumploadings`): its mid is multiplied by $\exp(\ell_{ik} Y)$, and the burst of Section 3.5 follows. A 1% market jump moves a stock of jump loading 1.5 by 1.5% and one of 0.4 by 0.4%. The stock's own jumps (Section 4.2) are separate: single-name news.

The common jumps carry a variance per day

$$J_i = \sum_k \lambda_k \ell_{ik}^2 s_k^2 \quad (28)$$

which is taken out of the stock's diffusion, so that `vol` remains the volatility of the close-to-close return. With ω the overnight share, β_b the break's share and m_d the day's volatility multiplier (Section 7),

$$(\sigma_i^d)^2 = m_d^2 \left[\sigma_i^2 (1 - \omega)(1 - \beta_b) - J_i N_{\text{days}} \right] \quad (29)$$

The correction follows the day's regime, so it averages J_i over the days and is positive on every day as soon as it is at composition; a stock whose common jumps would carry more variance than its day is refused.

8.3 Implied Correlation

With $V_i = \sigma_i^2 / N_{\text{days}}$ the day's variance, made of the overnight gap $G_i = \omega V_i$, the break's gap $B_i = (1 - \omega)\beta_b V_i$, the common jumps J_i and the diffusion $D_i = (1 - \omega)(1 - \beta_b)V_i - J_i$, the close-to-close returns of two stocks correlate at

$$\rho_{ij} = \frac{b_i \cdot b_j (\sqrt{G_i G_j} + \sqrt{B_i B_j} + \sqrt{D_i D_j}) + \sum_k \lambda_k \ell_{ik} \ell_{jk} s_k^2}{\sqrt{V_i V_j}} \quad (30)$$

which is $b_i \cdot b_j$ without common jumps and above it with them. For four US stocks on a market factor and three sectors, over two years of days:

Pair	Loadings	$b_i \cdot b_j$	Implied	Measured
NVDA, MSFT	same sector	0.545	0.62	0.61
NVDA, XOM	market only	0.358	0.46	0.50
NVDA, PG	market only	0.293	0.36	0.39
MSFT, XOM	market only	0.358	0.47	0.45
MSFT, PG	market only	0.293	0.37	0.38
XOM, PG	market only	0.248	0.33	0.32

Market loadings 0.65, 0.65, 0.55 and 0.45 for NVDA, MSFT, XOM and PG, sector loadings 0.35, 0.35, 0.45 and 0.30, jump loadings 1.5, 1.1, 0.8 and 0.4 on a market factor jumping 0.5 times a day with sizes of 1%. The standard error of a measured correlation is 0.04.

The close-to-close volatility of the four stocks over the same two years is 0.97 to 1.03 of the configured one, and the overnight gaps of NVDA and MSFT correlate at 0.53 against 0.545. Within the day two stocks correlate less than over a day: for NVDA and MSFT about 0.35 to 0.45 from five seconds to thirty minutes. The order-flow impact of Section 4.4 and the rounding to the tick add noise of the stock's own, which a day washes out.

8.4 Shared by Construction

The day's factor paths, gap normals and jumps are drawn from the factor seed of Section 2, which depends on the run's seed and the date alone, before the stock's own seed is set. Every stock of a date therefore shares them, with nothing passed from one stock to another; a stock's tape does not depend on which other stocks are simulated; an interrupted run resumed equals a full run; and a stock without loadings draws exactly what it would without factors, so its tape is unchanged to the bit.

9 Validation

Each module carries a suite of checks run by a k4unit harness, 48 for the shared configuration, 217 for simtick and 198 for simmarket, that hold the invariants (every trade inside its prevailing quote, spreads whole ticks, no trade or quote in a mid-day break, a stock's tape the same alone or with others) and the statistics below.

Property	NVDA	Reference for a US large cap
Arrivals vs analytic Hawkes mean, branching 0.3 and 0.9	within 1%	exact
Quote updates per trade	4.0	10 to 30 (set by <code>quotespertrade</code>)
Trades at the touch / midpoint / inside	80% / 12% / 8%	about 80 / 12 / 8
Trades outside the prevailing quote	0	0
Lag-1 autocorrelation of aggressor signs	0.40	0.3 to 0.5
Effective spread	0.49 bp	about 0.5 to 1 bp
Spread of one tick	81%	most of the day
Signed markout at 1 s / 1 min	0.34 / 0.17 cents	a fraction of a cent, reverting
Realized vol, 11:00 hour vs last hour	0.36 vs 0.60	U-shaped
Round lots / odd lots	35% / 55%	large mode at 100, odd lots over half
Off-exchange share	42%	about 40%
Close-to-close vol over two years, configured 0.45	0.449	0.45
Overnight share of daily variance, configured 0.3	0.29	0.3
Correlation of day regimes, configured 0.7	within 0.15	0.6 to 0.8
Daily correlation of NVDA and MSFT, implied 0.62	0.61	0.5 to 0.7 within a sector
Independence of two instruments' streams	within 0.03	0

Statistics reproduced by the implementation on the shipped configuration.

10 Limitations

- **Top of the book only:** there is no depth and no queue beyond the displayed size.
- **One-way link between the clocks:** trades seed quote updates; quotes do not excite trades.
- **Constant loadings:** the correlation between stocks does not rise on bad days and there is no leverage effect (a negative relation between returns and volatility).

- **Sessions:** the auction phases of the exchanges are single prints, there is no pre-market or after-hours trading, times are the exchange’s local time, and the tick size is one value per instrument, set for the price band the stock trades in.
- **Calendars:** the generator covers the NYSE, without special closures (days of mourning, disasters); other exchanges’ calendars are supplied as tables.

11 References

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- KDB-X Documentation, Modules. <https://code.kx.com/kdb-x/modules/module-index.html>
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A Example: One Day of Four US Stocks

This example generates one trading day, 23 September 2026, for four US stocks chosen to span the range of liquidity: a mega cap, a mid cap, a small cap and a micro cap. The less liquid the stock, the wider its spread, the fewer its trades and the larger a given trade is against its normal activity.

Stock	Tier	Listing	Price	Market cap	Daily volume (shares)
The Coca-Cola Company (KO)	mega cap	NYSE	\$78	\$335 bn	12–17 m
Hasbro, Inc. (HAS)	mid cap	Nasdaq	\$90	\$12 bn	1.7–2.6 m
Sally Beauty Holdings, Inc. (SBH)	small cap	NYSE	\$16	\$1.5 bn	1.1–1.5 m
Vera Bradley, Inc. (VRA)	micro cap	Nasdaq	\$3.75	\$0.1 bn	0.15–0.5 m

Approximate figures from quotes during 2026.

Calibration. Price, listing and daily share volume follow recent quotes for each stock; volatility, spread, trade sizes and venue shares are estimates for stocks of each type and should be checked against market data before any quantitative use. `tradesperday` and `avgqty` are set together so that the number of trades times the mean trade size of the mixture (Section 6.3) is close to each stock’s daily share volume. `impactticks` follows the square-root law, $\kappa \delta = Y \sigma_{\text{day}} \sqrt{Q/V} S_0$ with $Y = 0.3$: small in ticks for the low-priced stocks, but largest in basis points for them, about 3.4 bp per trade for VRA against 0.1 bp for KO.

Venues. Stocks trade most on their own listing exchange, so the lit venue shares depend on the listing: the NYSE split for KO and SBH, the Nasdaq split for HAS and VRA. They apply to the trades printed on exchange; `offexchangeshare` of all trades print off-exchange (venue TRF), and the auctions print on `primaryvenue`. The resulting share of all trades by venue is:

Venue	Coca-Cola	Hasbro	Sally Beauty	Vera Bradley
TRF (off-exchange)	42.0%	40.0%	45.0%	50.0%
XNAS	12.8%	24.0%	12.1%	20.0%
XNYS	16.2%	3.0%	15.4%	2.5%
ARCX	8.1%	9.0%	7.7%	7.5%
BATS	7.0%	7.8%	6.6%	6.5%
EDGX	7.0%	7.8%	6.6%	6.5%
IEXG	2.9%	3.6%	2.8%	3.0%
MEMX	4.1%	4.8%	3.9%	4.0%

Seeds. Each stock gets its own seed. With a shared seed the four would draw the same random numbers and their price paths would move in step. The four stocks carry no factor loadings, so they are independent; with loadings they would be run under one seed by `di.simmarket`, which gives each stock its own stream and all of them the day's factors (Section 8).

All parameters. The table lists the 83 configuration keys that trades and quotes depend on, with the names used in the code. A value spanning the four columns is shared by all four stocks; it comes from the market file `us_largecap` and the `normal` scenario unless the instrument row overrides it. `baseintensity` is not set but derived by `compose` from `tradesperday` (Section 3.2), here with a mean profile level of 1.004 and no jump bursts.

Parameter	The Coca-Cola Company	Hasbro, Inc.	Sally Beauty Holdings, Inc.	Vera Bradley Inc.									
<i>Essential (instrument)</i>													
sym	KO	HAS	SBH	VRA									
price	78.00	90.00	16.00	3.75									
drift			0.05										
vol	0.16	0.30	0.45	0.65									
tradesperday	100000	18000	6000	1300									
<i>Session (market)</i>													
openingtime			09:30										
closingtime			16:00										
breakstart		none (no mid-day break)											
breakend			none										
tradingdays			252										
rngmodel			pseudo										
ticksize			0.01										
printgrid			0.1										
clock			transaction										
<i>Arrivals (market)</i>													
alpha			0.3										
beta			1.0										
profile	1.6	1.2	1.0	0.85	0.75	0.7	0.65	0.7	0.75	0.85	1.0	1.2	1.8
jumpburst			3000	(no effect: no jumps)									
jumpburstminutes			1.0	(no effect: no jumps)									
quotespertrade													4
quotetradelink													0.5
<i>Auctions (market)</i>													
openauctionpct				0.01									
closeauctionpct				0.08									

Parameter	The Coca-Cola Company				Hasbro, Inc.				Sally Beauty Holdings, Inc.				Vera Bradley, Inc.				
breakauctionpct	0 (no effect: no break)																
Factors (market)																	
factors	market				Technology		Energy		Consumer		Staples						
factorprofile	1.6	1.2	1.0	0.85	0.75	0.7	0.65	0.7	0.75	0.85	1.0	1.2	1.8				
factorjumpintensities					0.5				0	0	0						
factorjumpvols					0.01				0	0	0						
Factor loadings (instrument, optional)																	
factorloadings	not set: the four stocks take nothing from the factors																
jumploadings	not set																
Trade sizes (market, instrument overrides)																	
qtypodel									mixture								
avgqty	40				40								150		200		
qtyvol									0.9								
roundlotshare	0.35				0.25								0.35		0.35		
roundlots					100		200	300	500	1000							
roundlotweights					0.5		0.2	0.1	0.12	0.08							
blockshare									0.002								
blockqty	10000				5000								5000		5000		
blockqtyvol									0.5								
Quotes (market, instrument overrides)																	
spreadticks	1.05				2.5								1.5		2		
spreadopenmult									2.5								
spreadmidmult									1.0								
spreadclosemult									0.9								
spreaddecayminutes									15								
spreadactivity									0.5								
activitywindowseconds									60								
activityclip									0.25		4						
avgquotesize	2000				300								1000		800		
quotesizevol									0.6								
quotelot									100								
imbalance signal									0.15								
Trades against the quotes (market, instrument overrides)																	
sidepersistence									0.7								
midpointshare									0.12								
improvementshare									0.08								
improvementtick									0.1								
offexchangeshare	0.42				0.40								0.45		0.50		
offexchangeinside									0.95								
venues	XNAS				XNYS	ARCX	BATS	EDGX	IEXG	MEMX							
venueshares	0.22	0.28	0.14		0.40	0.05	0.15		0.22	0.28	0.14		0.40	0.05	0.15		
	0.12	0.12	0.05		0.13	0.13	0.06		0.12	0.12	0.05		0.13	0.13	0.06		
	0.07				0.08				0.07				0.08				
primaryvenue	XNYS				XNAS				XNYS				XNAS				
Order-flow impact (market, instrument overrides)																	
impactticks	0.07				0.4								0.2		0.15		
impacthalflifeseconds									30								
impactpermanent									0.3								
Scenario: normal																	
volmult									1.0								
volumemult									1.0								
spreadmult									1.0								
pricemodel									gbm								
jumpintensity									0								
jumpmean									0								
jumpvol									0								

Parameter	The Coca-Cola Company	Hasbro, Inc.	Sally Beauty Holdings, Inc.	Vera Bradley, Inc.
<i>Scenario: day-to-day keys (used by di.simmarket only)</i>				
overnightshare		0.3		
breakshare		0.05 (no effect: no break)		
gapdayweight		0.25		
regimepersistence		0.7		
regimecorr		0.7		
volregimesd		0.3		
volumeregimesd		0.3		
<i>Run</i>				
tradingdate		2026.09.23		
seed	42	43	44	45
generatequotes		true		
<i>Derived by compose</i>				
baseintensity	2.98	0.536	0.179	0.0387
factorseed	2087121688	1277087489	1963630985	790960975
jumpcomp		1		

Running the example. In q, with the shipped files loaded as `market` and `sc` (the normal scenario), each stock is a dictionary of the keys it sets or overrides, composed and run with its own seed:

```
nas:`XNAS`XNYS`ARCX`BATS`EDGX`IEXG`MEMX!0.40 0.05 0.15 0.13 0.13 0.06 0.08
nys:`XNAS`XNYS`ARCX`BATS`EDGX`IEXG`MEMX!0.22 0.28 0.14 0.12 0.12 0.05 0.07
k:`sym`primaryvenue`price`drift`vol`tradesperday`spreadticks`avgqty,
  `roundlotshare`blockqty`avgquotesize`offexchangeshare`impactticks
ins:flip k!(`KO`HAS`SBH`VRA; `XNYS`XNAS`XNYS`XNAS;
  78 90 16 3.75; 4#0.05; 0.16 0.30 0.45 0.65;
  100000 18000 6000 1300; 1.05 2.5 1.5 2; 40 40 150 200;
  0.35 0.25 0.35 0.35; 10000 5000 5000 5000; 2000 300 1000 800;
  0.42 0.40 0.45 0.50; 0.07 0.4 0.2 0.15)
ins:update venues:4#enlist key nas,
  venueshares:(value nys;value nas;value nys;value nas) from ins
run1:{[i;s] d:`tradingdate`seed!(2026.09.23;s);
  simtick.run simtick.compose[market;i;sc;d]}
r:run1'[ins;42 43 44 45]
trade:`sym`time xasc raze r@\:`trade
quote:`sym`time xasc raze r@\:`quote
```